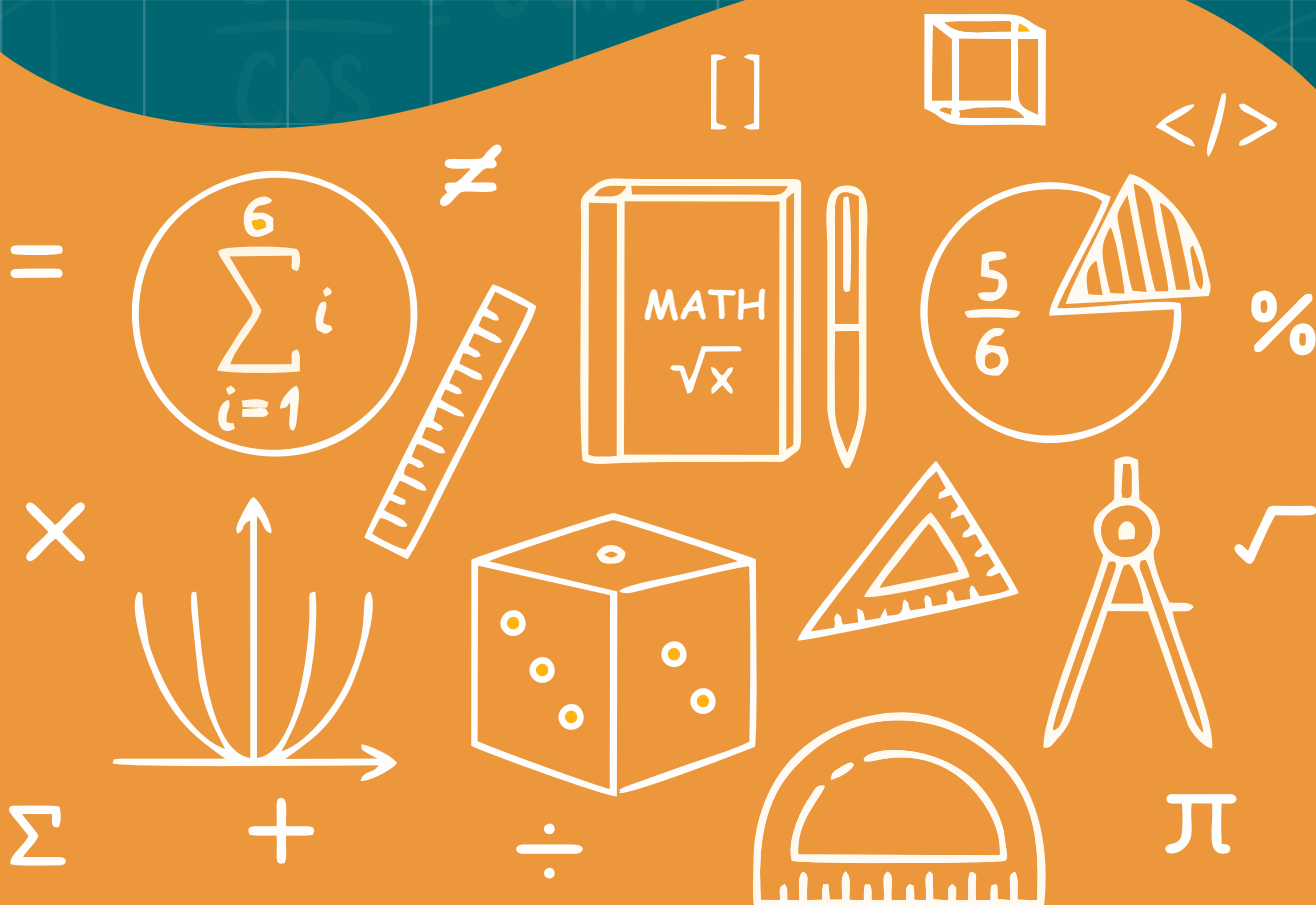


UNIT 11

PERIMETER AND AREA



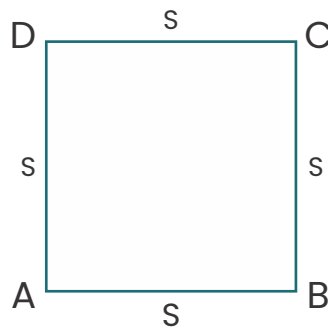
PERIMETER

Perimeter is the length of boundary of a plane closed figure and area is the measurement of surface of plane closed figure.

Perimeter of square:

We know that the sum of measures of all the sides of a square is its perimeter.

In the figure, the length of each side of square is S units.



Perimeter Sum of all sides

$$\begin{aligned}\text{So, Perimeter} &= S + S + S + S \\ &= 4S\end{aligned}$$

i.e. Perimeter of a square = 4 x Side

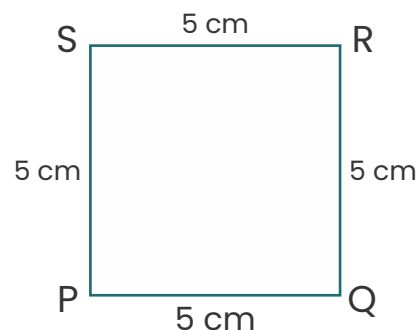
Where S denotes the length of a side.

Example 1. Find the perimeter of a square whose each side is 5 cm long.

Solution:

Here side = 5 cm

$$\begin{aligned}\text{Now Perimeter} &= 4 \times \text{side} \\ &= 4 \times 5 \\ &= 20 \text{ cm}\end{aligned}$$



$$\text{Side of a square} = \frac{\text{Perimeter of a square}}{4}$$

Example 2. Find the length of side of a square whose perimeter is 40 cm.

Solution: Here perimeter = 40 cm

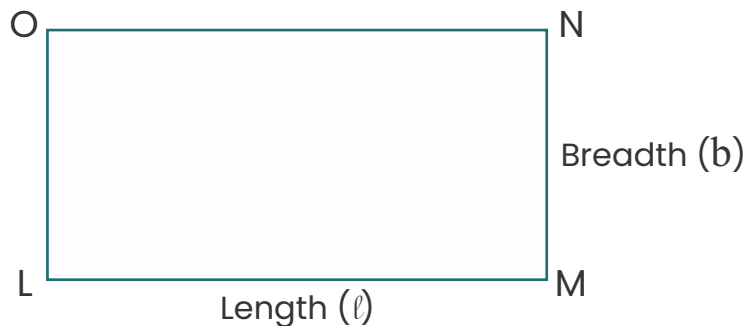
We know that, side of a square = $\frac{\text{Perimeter of square}}{4}$

$$= \frac{40}{4} = 10 \text{ cm}$$

So, length of each side is **10 cm**.

Perimeter of a rectangle:

The sum of the measures of all sides of a rectangle is called its perimeter.



$$\text{Perimeter} = \text{Length} + \text{Breadth} + \text{Length} + \text{Breadth}$$

$$= l + b + l + b$$

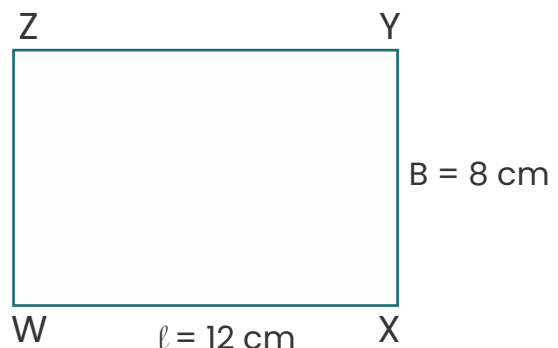
$$= 2l + 2b = 2(l + b)$$

So, Perimeter of rectangle = $2(l + b)$

Example 3. Find the perimeter of rectangle whose length and breadth are 12 cm and 8 cm respectively.

Solution: Here $l = 12\text{cm}$
 $b = 8\text{cm}$

$$\begin{aligned} \text{Perimeter of rectangle} &= 2(l + b) \\ &= 2(12 + 8) \\ &= 2(20) \\ &= 40\text{cm} \end{aligned}$$



So, the required perimeter of a given rectangle is **40 cm**.

Example 4. Find the length of rectangle whose breadth is 6 cm and its perimeter is 28 cm.

Solution: Here $\ell = ?$

$$b = 6 \text{ cm}$$

Perimeter of a rectangle = 28 cm

Now Perimeter of rectangle = $2(\ell + b)$

$$\text{i.e. } 28 = 2(\ell + 6)$$

$$\text{or } \frac{28}{2}(\ell + 6)$$

$$\text{or } 14 = \ell + 6$$

$$\text{or } 14 - 6 = \ell$$

$$\text{or } 8 \text{ cm} = \ell$$

$$\text{or } \ell = 8 \text{ cm}$$

So, the required length is **8 cm**.

Area of a square:

We know that:

$$\text{Area of a square} = \text{side} \times \text{side}$$

Area of rectangle:

We know that:

$$\text{Area of rectangle} = \text{length} \times \text{breadth}$$

Example 5. Find the area of a square whose each side is 3 cm.

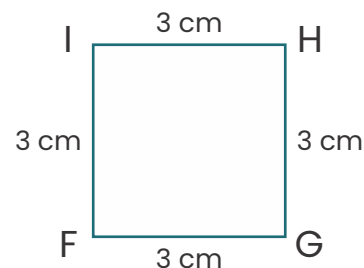
Solution: Here Side = 3 cm

We know that

Area of square = Side x Side

$$= 3 \times 3$$

$$= 9 \text{ sq. cm}$$

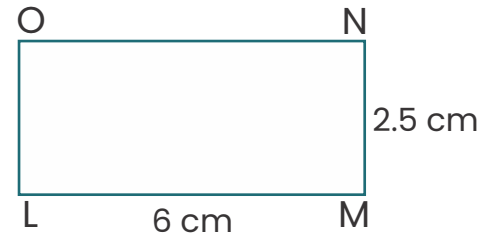


Example 6. Find the area of a rectangle in which length is 6 cm and breadth is 2.5 cm.

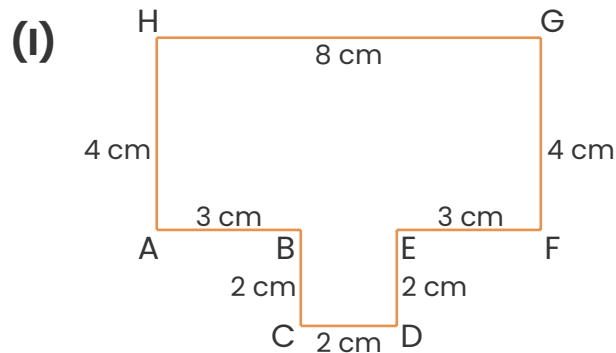
Solution: Here Length = 6 cm

Breadth = 2.5 cm

Now Area = length $\times$ breadth = 6×2.5
 $= 15.0$ sq. cm



Example 7. Find area and perimeter of the following figures.



Solution:

Perimeter of figure

According to the figure

Perimeter = $8 + 4 + 3 + 2 + 2 + 2 + 3 + 4 = 28$ cm

Area of figure

In the figure there is a rectangle and square.

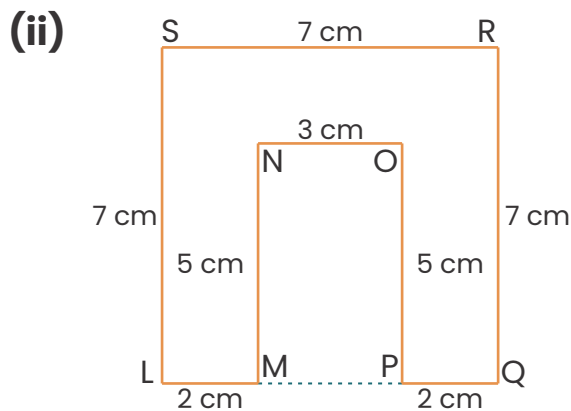
According to the figure:

Area of figure = Area of rectangle + Area of square

$$= 8 \times 4 + 2 \times 2$$

$$= 32 + 4$$

$$= 36 \text{ sq. cm}$$



Solution:

Perimeter of figure

According to the figure:

$$\text{Perimeter} = 7 + 7 + 7 + 2 + 5 + 3 + 5 + 2 = 38 \text{ cm}$$

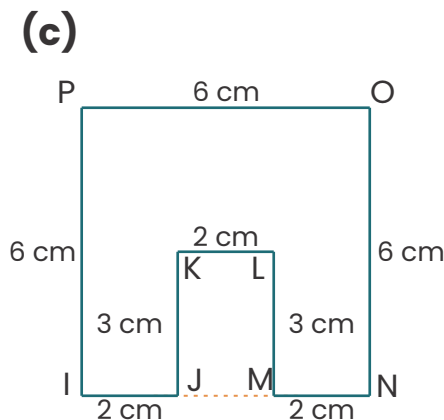
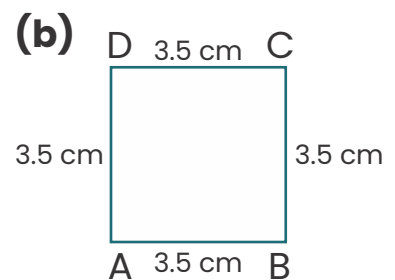
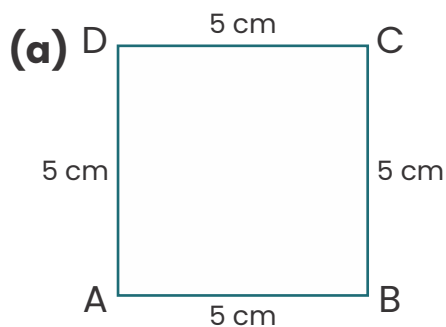
Area of figure

According to the figure

$$\begin{aligned} \text{Area of figure} &= \text{Area of square} - \text{Area of rectangle} \\ &= 7 \times 7 - 3 \times 5 \\ &= 49 - 15 \\ &= 34 \text{ sq. cm} \end{aligned}$$

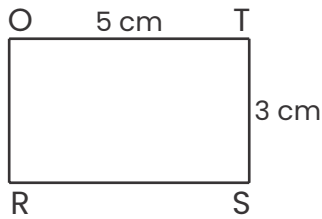
EXERCISES 11.1

Q1. Find the perimeter and area of each of the following figures.

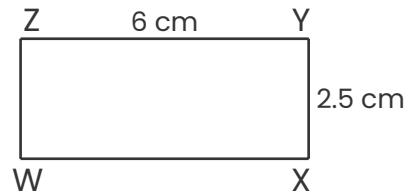


Q2. Find the perimeter and area of the following rectangles.

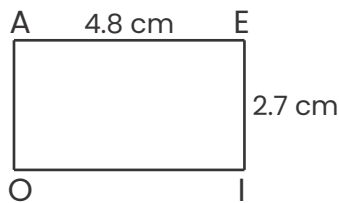
(a)



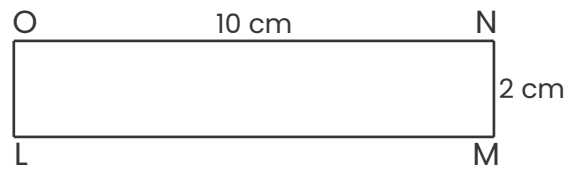
(b)



(c)



(d)



Q3. Find the perimeter and area of:

(i) Rectangle with sides 5 cm, 2.5 cm, 5 cm and 2.5 cm

(ii) Square with each side 5 cm.

Q4. Find the side of square whose perimeter is:

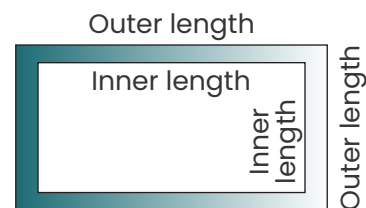
(i) 48 cm

(ii) 50 cm

AREA

Find area of path (inside or outside) of a rectangle or square

In the adjacent figure shaded portion represents the path.



Here

Area of the large rectangle (including path)

= Outer length x outer breadth

Also area of the small rectangle (excluding path)

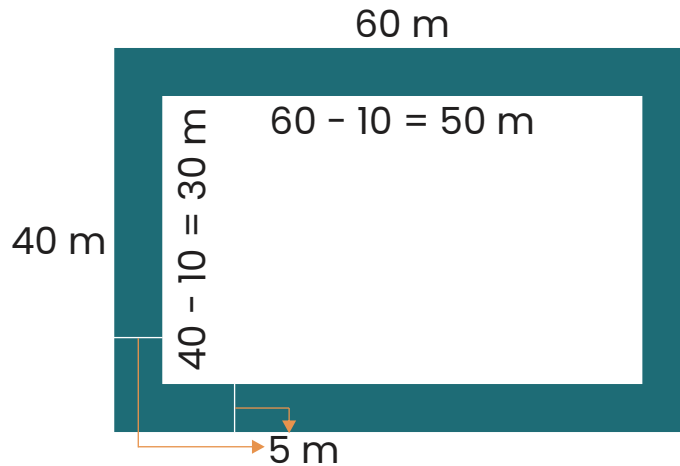
= Inner length x Inner breadth

So,

$$\text{Area of path} = \text{Area of large rectangle} - \text{Area of small rectangle}$$

Example. A field is 60 m long and 40 m wide. A path 5 m wide is constructed all around it inside the field. Find the area of the path.

Solution:



Area of large rectangle

$$= \ell \times b$$

$$= 60 \times 40$$

$$= 2400 \text{ sq. m}$$

Area of small rectangle

$$= \ell \times b$$

$$= 50 \times 30$$

$$= 1500 \text{ sq. m}$$

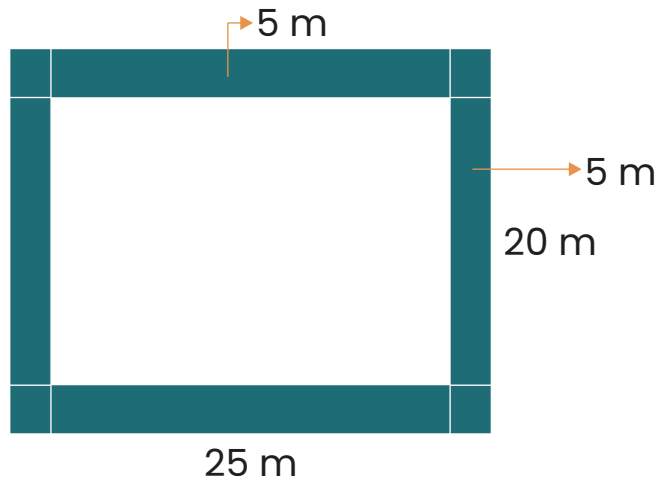
So, Area of path = 2400 - 1500

$$= \mathbf{900 \text{ sq. m}}$$

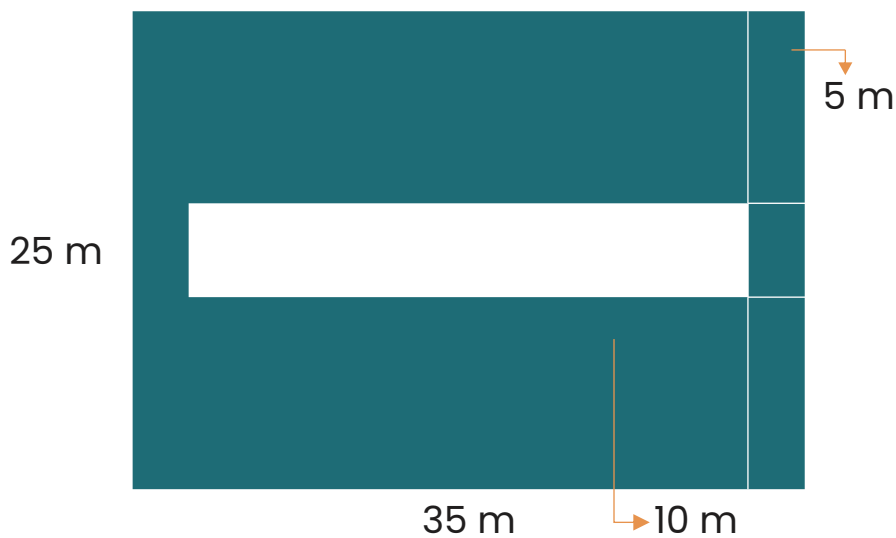
EXERCISE 11.2

Q1. Find the area of shaded portion in the following figures.

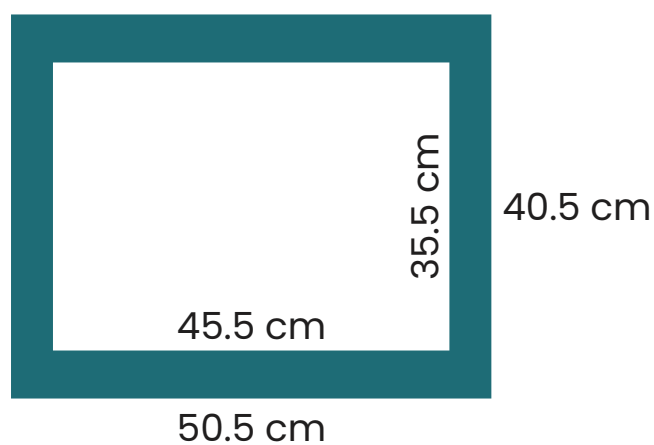
(i)



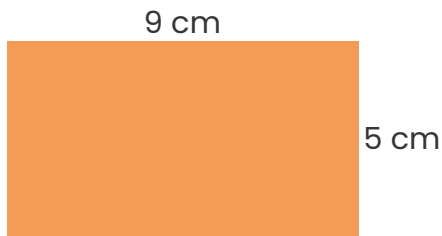
(ii)



(iii)



Q2. Calculate the area and perimeter of the following figures.



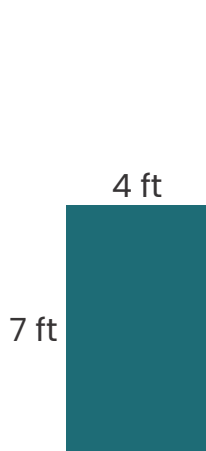
Perimeter _____ cm
Area _____ cm²



Perimeter _____ in
Area _____ in²



Perimeter _____ cm
Area _____ cm²



Perimeter _____ ft
Area _____ ft²

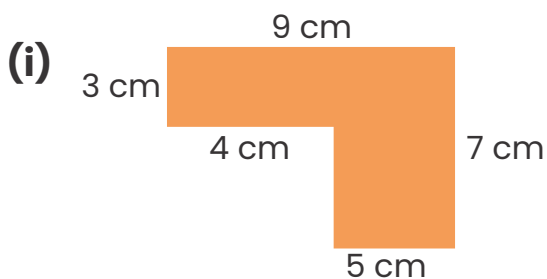


Perimeter _____ ft
Area _____ ft²

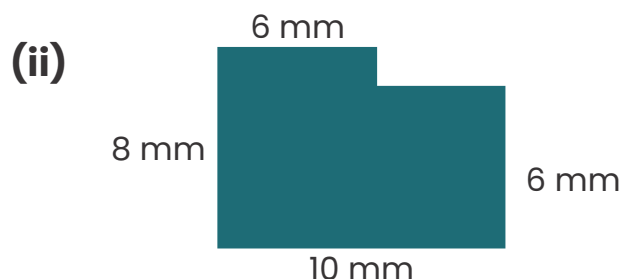


Perimeter _____ ft
Area _____ ft²

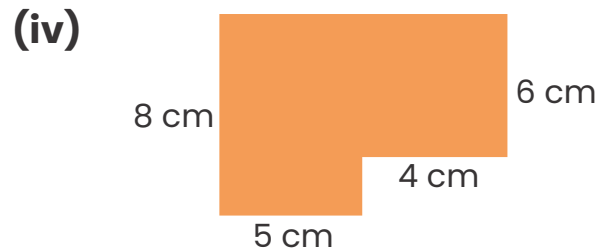
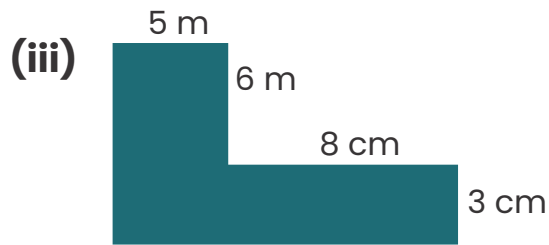
Q3. Find the area and perimeter of the following:



Area = _____ square cm (cm²)
Perimeter = _____ m



Area = _____ square mm (mm²)
Perimeter = _____ mm



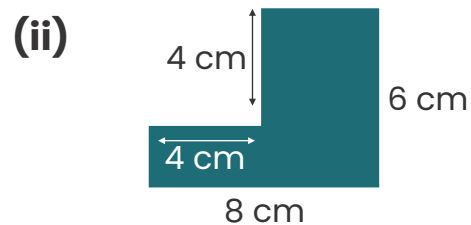
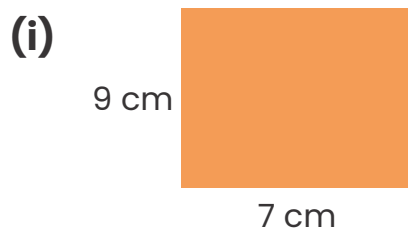
Area = _____ square cm (m^2)

Perimeter = _____ m

Area = _____ square mm (cm^2)

Perimeter = _____ cm

Q4. Calculate the perimeter and area:

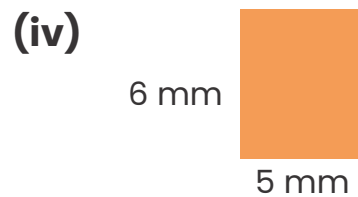
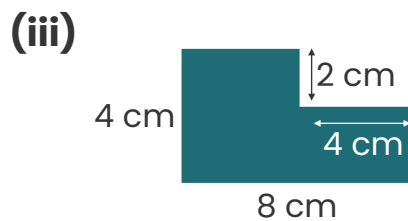


Perimeter: _____

Area: _____

Perimeter: _____

Area: _____



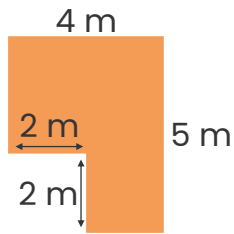
Perimeter: _____

Area: _____

Perimeter: _____

Area: _____

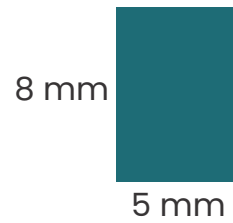
(v)



Perimeter: _____

Area: _____

(vi)



Perimeter: _____

Area: _____

WORD PROBLEMS

Q1. Find the cost of fencing a square shaped park of side 250 m at the rate of Rs 30 per metre.

Q2. Find the cost of fencing a rectangular park of length 200 m and breadth 150 m at the rate of Rs 30 per metre.

Q3. Find the length of a plot of width 15 m whose area is 615 sq. m.